ITA0502-COMPUTER VISION

Activity-1

1. Medical Image Enhancement

A doctor receives an X-ray image in which the bones are visible, but the darker tissues have very poor visibility.

Questions:

1. Which gray-level transformation would you recommend?
2. Why is it suitable?
3. What effect will it have on the image?

2. Night-Time Surveillance

A CCTV camera captures a parking lot at night. Most pixel values are concentrated between **0 and 40**, making the image very dark.

Questions:

1. Which transformation would improve the visibility?
2. Why is it more suitable than negative transformation?
3. Predict the appearance of the enhanced image.

3. Fingerprint Recognition

A fingerprint image contains many fine ridge details, but the contrast between the ridges and valleys is very low.

Questions:

1. Which gray-level transformation would you apply?
2. Explain why.
3. What would happen if a log transformation were applied instead?

Activity

1. Medical Image Enhancement:

1. Recomended Gray-level Transformation:

Power-law (Gamma) Transformation (specifically with $\gamma < 1$) or log transformation.

2. Why it is suitable:-

In this X-ray. the dark tissue regions correspond to low pixel intensity values, while the bright bone structures correspondi[ng] high intensity values. A power law transform with $\gamma < 1$ (or log transformation) maps a na[rrow] range of dark input intensity values effe[ctively] strecthing and brighting the details hidden in the dark region.

3. Effect on the image:-

The overall contrast in the dark tissue area will be significantly enhanced, making ~~stro~~soft tissue structure.

2. Night - Time Surveillance:-

1. Recommended Transformation:-

Histogram Equivalization or contrast stretching /log transformation.

2. Why it is more suitable than negative transforma[tion]

Negative transformation only inverts colors (turning light dark and dark light).

Histogram equivalization actually expands the concentrated dark pixel values (0-40) accross the full 0-255 range to increase low contrast.

3. Predicted Appearance:-

A brighter, clearer image where ~~th~~ vehicles people and objects in the parking lot are easily distinguishable.

3. Fingerprint recognition:-

1. Recomenended transformation

Contract streching (or Threshholding).

2. Why:- It expands the narrow range of low contrast intensity values accross the entire scale, creating sharp separation b/w ridges and valleys.

3. Effect of log transformation:-

It would overly brighten the image and compress light values, washing out valley details and reducing overall ridge contrast.